CSE1142 Computer Programming II Midterm Exam KEY (Spring 2021)

Q1. (21 pts.)

a) (9 pts.) What is printed as a result of executing the code segment?

```
ArrayList<String> animals = new ArrayList<>();
animals.add("dog");
animals.add("cat");
animals.add("snake");
animals.set(2, "tiger");
animals.add(1, "fish");
animals.remove(3);
System.out.println(animals);
```

Answer: [dog, fish, cat] // Each 3 pts.

b) (12 pts.) Show the output of the following program:

```
public class UnderGrad extends Student {
  public static void main(String[] args) {
    new UnderGrad();
  }
  public UnderGrad() {
    this("UnderGrad: A");
    System.out.println("UnderGrad: B");
  }
  public UnderGrad(String s) {
    System.out.println(s);
  }
}
class Student extends Person {
  public Student () {
    System.out.println("Student");
  }
}
class Person {
  public Person() {
    System.out.println("Person");
  }
```

Answer: Person

Student

UnderGrad: A

UnderGrad: B

//Each 3 pts.

Q2. (29 pts.)

```
class Point implements Comparable<Point>{
  private double x;
  private double y;

  public Point(double x, double y) {
    this.x = x;
    this.y = y;
  }

  // 5pts
  public boolean equals(Object o) {
        return x == ((Point)o).x && y == ((Point)o).y;
  }
  // 5pts
  public int compareTo(Point p) {
          if (x > p.x) {
              return 1;
            }
            else if (x < p.x) {
              return -1;
            }
            else if (y > p.y) {
              return 1;
            }
            else if (y == p.y) {
              return 0;
            }
            else {
              return -1;
            }

  }
  // 5pts
  public static Point max(Point p1, Point p2) {
        if(p1.compareTo(p2)>=0)
              return  p1;
        else
              return p2;
  }

  public static void main(String[] args) {
    // 8 pts
    // Create a Point object with x=3 and y=4
    Object p1 = new Point(3, 4);
    // Create a Point object with x=3.4 and y=1.4
    Object p2 =new Point(3.2,1.4);

    //invoke the corresponding methods
    System.out.println("p1 is equal to p2? "+p1.equals(p2));
    System.out.println("p1 compared to p2 " + ((Point)p1).compareTo((Point)p2));
    System.out.println("p2 compared to p1 " + ((Point)p2).compareTo((Point)p1));
  }
}
```

```
Output:  p1 is equal to p2? False   //Each 2 pts.
         p1 compared to p2 -1
         p2 compared to p1 1
```

Q3. (16 pts.) Consider the following program:

```
public static void main(String args[]) {
    try {
      m1();
    }
    catch (Exception1 ex1) {
      statement1;
    }
    catch (Exception2 ex2) {
      statement2;
    }
    statement3;
}
public void m1() {
    try {
      statement4;
      statement5;
    }
    catch (Exception1 ex1) {
          statement6;
    }
    finally {
          statement7;
    }
    statement8;
}
```

Answer the following questions:

a) If no exception occurs, which statements (**statement1, …, statement8**) will be executed in what order?
// 5pts.
Answer: Statements 4, 5, 7, 8, 3

b) If **statement4** causes an exception in type of **Exception1**, which statements (**statement1, …, statement8**) will be executed in what order?
// 5pts.
Answer: Statements 4, 6, 7, 8, 3

c) If **statement4** causes an exception in type of **Exception2**, which statements (**statement1, …, statement8**) will be executed in what order?
// 4pts.
Answer: Statements 4, 7, 2, 3

d) If **statement4** causes an exception in type of **Exception3**, which statements (**statement1, …, statement8**) will be executed in what order?
// 2pts.
Answer: Statements 4, 7

Q4. (34 pts.) Consider that the following objects are created based on the class hierarchy below:

```
LocationControl obj1 = new SmartLight();
Programmable obj2 = new SmartPlug();
MotionControl obj3 = new SmartCamera();
SmartObject obj4 = new SmartPlug();
```

a) Answer the following questions (**True** / **False**): **//Each 2 pts.**

1) Is **obj1 instanceof Programmable** ? TRUE
2) Is **obj2 instanceof SmartObject** ? TRUE
3) Is **obj3 instanceof LocationControl** ? FALSE

b) Suppose the method **method2** is defined in the **SmartPlug.** State the execution result (**compile error**, **runtime error** - **ClassCastException**, or **no problem**) for the following statements and explain the reason for the error cases. Each case is independent from each other. **//Each 2 pts.**

1) `Object obj5 = (SmartObject)obj1;`
   NO PROBLEM

2) `SmartCamera obj6 = (SmartCamera)obj2;`
   Runtime Error: Class Cast Exception

3) `obj2.method2();`
   Compile Time Error -> Casting required

4) `MotionControl obj7 = (SmartCamera)obj3;`
   NO PROBLEM

5) `((SmartPlug)obj4).method2();`
   NO PROBLEM

c) Suppose the abstract method **`method1()`** is defined in the **Programmable**. What class(es) must implement this method?
SmartLight and SmartPlug **// Each 2 pts.**

d) Implement a Test class, define a main method that creates an **`ArrayList`** of **`SmartObjects`** of two **`SmartPlugs`**, two **`SmartLight`**, and one **`SmartCamera`**, then iterate over the list and invoke the **`method1()`** for the corresponding objects. **//14 pts**

```
ArrayList<SmartObject> myobjects = new ArrayList<>();  // 1pnt
myobjects.add(new SmartPlug()); // 1pnt
myobjects.add(new SmartPlug()); // 1pnt
myobjects.add(new SmartLight()); // 1pnt
myobjects.add(new SmartLight()); // 1pnt
myobjects.add(new SmartCamera()); // 1pnt
for(int i=0; i<myobjects.size(); i++) { // 3 pts
```

```
            if(myobjects.get(i) instanceof Programmable) // 2 pts
                ((Programmable)myobjects.get(i)).method1(); //3pts.
}
```